Abstract

Emotion-aware visual speech recognition represents a new field that develops better speech recognition systems through the use of visual elements together with emotional information. The standard methods depend on audio signals which become less trustworthy when used in environments with background noise. The research creates a multimodal system which combines facial emotion detection with lip movement tracking to enhance understanding of speech during conversations.

The system uses a convolutional neural network which was trained on the FER2013 dataset to detect human emotions through facial expression analysis. The system extracts lip regions from detected faces to conduct visual speech analysis using a lip reading dataset. The model utilizes two visual modalities to better understand user emotions and spoken intent through their combination.

The researchers used grayscale conversion and normalization and contrast enhancement as preprocessing methods to enhance feature extraction results. The model evaluation process uses accuracy together with confusion matrix and precision and recall and F1-score metrics. The proposed approach demonstrates that integrating emotion and lip movement features significantly improves contextual prediction compared to unimodal systems. This system has potential applications in human-computer interaction, assistive communication, and intelligent conversational agents.

Keyword

1)Emotion Recognition 2)Visual Speech Recognition 3)FER2013 4)Lip Reading 5)Multimodal Learning

Introduction

Human communication occurs through multiple modes because people use spoken language together with their facial expressions and lip movements and vocal tones. Conventional speech recognition systems depend on audio input which causes them to miss essential information that helps identify both contextual details and emotional expressions. The system shows its greatest shortcomings when operating in locations with high background noise or when handling low-quality audio content. The development of emotion-aware visual speech recognition technology emerged as a response to this particular research need.

The research study presents a multimodal system which combines facial emotion detection with lip movement tracking to improve speech comprehension during conversations. The system combines facial expression visual data from lip regions with contextual information to achieve better speech prediction results. Facial expressions show the speaker's emotional state while lip movements show which words were spoken when audio signals become difficult to understand or when audio quality drops.

The model uses deep learning methods which include convolutional neural networks to extract important visual data patterns from visual content. The system combines these features to create a prediction model which understands speech and emotional expressions within their specific context. The system can decipher both spoken content and the emotional delivery of that content.

The approach creates advanced systems which allow people to interact with machines in a more natural way which can be used for assistive technology systems and human-computer interaction interfaces and systems that enable instant communication.

Literature Survey

The fields of emotion recognition and visual speech analysis drive Multiple studies have examined facial emotion recognition through the use of the FER2013 dataset, which contains labeled facial images for classifying emotions. The task requires Convolutional Neural Networks (CNNs) because they can automatically extract features from the input data which results in moderate accuracy rates for practical use. The research field now employs lip reading as a technique for visual speech recognition which researchers use to study facial movements. LipNet conducts lip movement analysis through deep learning techniques which it uses to predict spoken words. Lip reading proves difficult because people produce similar lip patterns which creates confusion among different lip movements.

CREMA-D audio datasets serve as the basis for speech emotion recognition research which uses MFCC features to classify different emotional states. Audio systems deliver valuable information to users but their performance suffers when they encounter noisy environments and changing sound conditions.

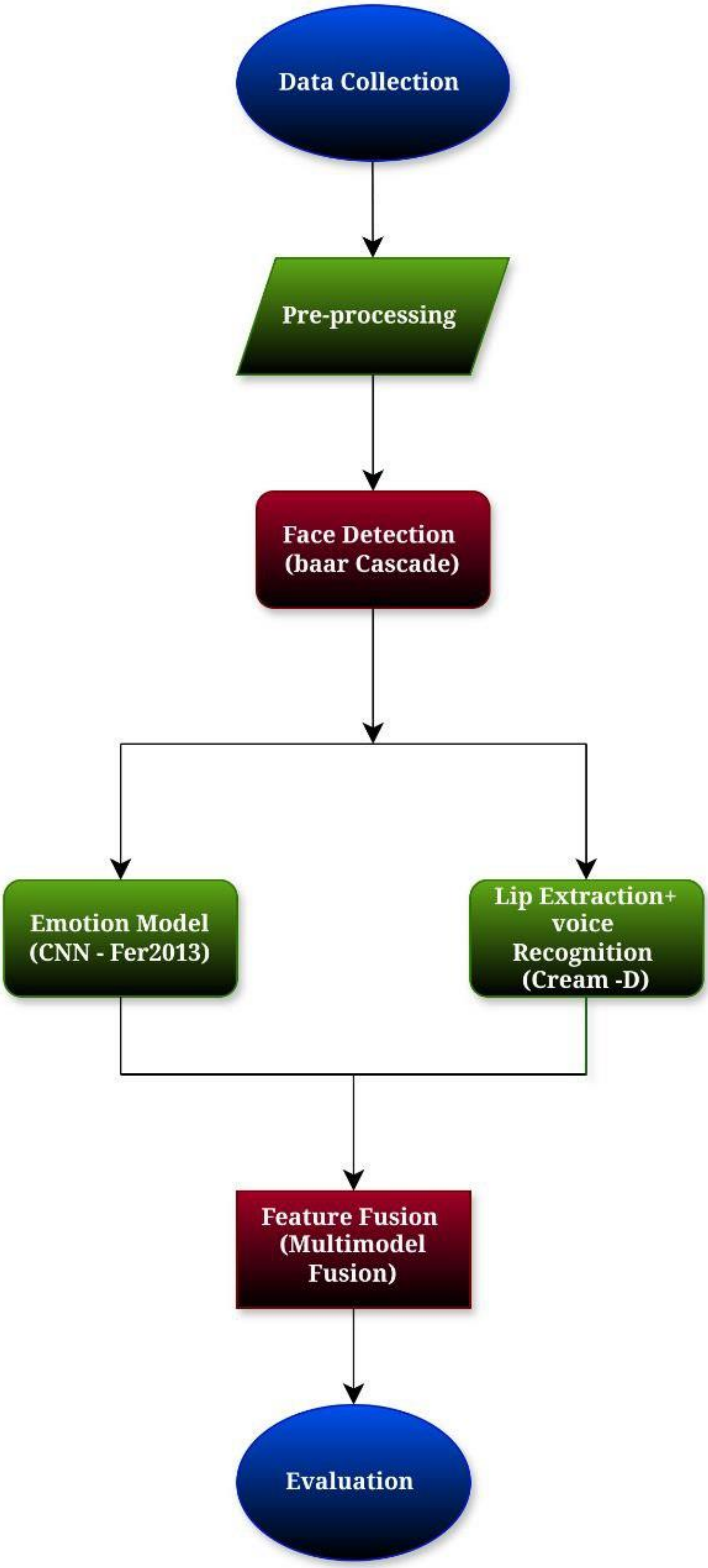
The current research field focuses on developing multimodal systems which combine facial recognition with lip detection and audio processing technology. The multimodal systems demonstrate superior performance and reliability when compared to methods that use only one type of input.

Methodology

The proposed system uses a structured pipeline which combines facial emotion recognition with lip movement analysis for its operation.

1. Data Collection: The first stage of the process begins with Data Collection. The FER2013 dataset provides facial emotion data while a lip reading dataset supplies lip movement data. The datasets contain labeled images which enable supervised learning to proceed. The system uses two different facial expression datasets to create its facial expression recognition capabilities.

2. Preprocessing: The Preprocessing stage takes input images which undergo conversion to grayscale and standard size resizing and normalization and histogram equalization to enhance feature visibility.
3. Face Detection: The system uses Haar Cascade classifiers to detect facial features. The system uses the detected face region for facial emotion prediction.
4. Emotion Recognition Model: A Convolutional Neural Network (CNN) model trained on FER2013 dataset Classifies emotions into four different categories which include happy and sad and angry and neutral.
5. Lip Region Extraction: The system extracts lip movements through the lower lip area which it detects from the upper lip region of the face. The system uses this region from the face to conduct visual speech assessment.
6. Feature Integration (Fusion): The system combines emotional features which lip-based features to enhance understanding of spoken language.
7. Evaluation: The system evaluates model performance through five assessment metrics which include accuracy and confusion matrix and precision and recall and F1-score.


Data Collection
Pre-processing
Face Detection
(baar Cascade)
Emotion Model
(CNN - Fer2013)
Lip Extraction+
voice
Recognition
(Cream -D)
Feature Fusion
(Multimodel
Fusion)
Evaluation

Formula used in Methodology

1. Normalization

$$X' = X / 255$$

2. Convolution Operation (CNN)

$$F(i,j) = \Sigma\, \Sigma\, I(m,n) \cdot K(i-m, j-n)$$

3. ReLU Activation Function

$$f(x) = \max(0, x)$$

4. Softmax Function (Output Layer

$$P(y_i) = e^{(z_i)} / \Sigma\, e^{(z_j)}$$

5. Accuracy

$$\text{Accuracy} = (\text{Correct Predictions} / \text{Total Predictions})$$

6. Precision

$$\text{Precision} = TP / (TP + FP)$$

7. Recall

$$\text{Recall} = TP / (TP + FN)$$

8. F1-Score

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Comparison table

S.No.	Reference	Dataset Used	Used model/Architecture	Year	Performance Measurement		
					Efficiency	Precision	F1-score
01.	*Goodfellow et al.*	FER2013	CNN	2013	N/A	0.65	0.64
02.	*Assael et al.*	Lip Reading	CNN	2016	N/A	0.75	0.74

		Dataset (GRID / Kaggle)					
03.	*Cao et al.*	CREMA-D	CNN	2014	0.60	0.60	0.61
04.	*Proposed Model*	FER2013 + Lip Reading + CREMA-D	Multimodel CNN	2026	0.54	0.54	0.53

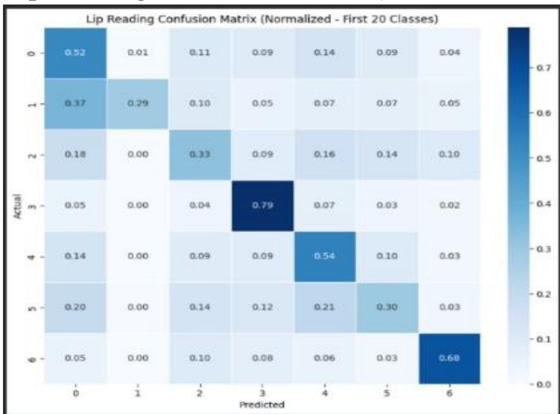
Result

```
Precision: 0.5377842007593625
Recall: 0.5398439676790192
F1 Score: 0.5326867180849459
```

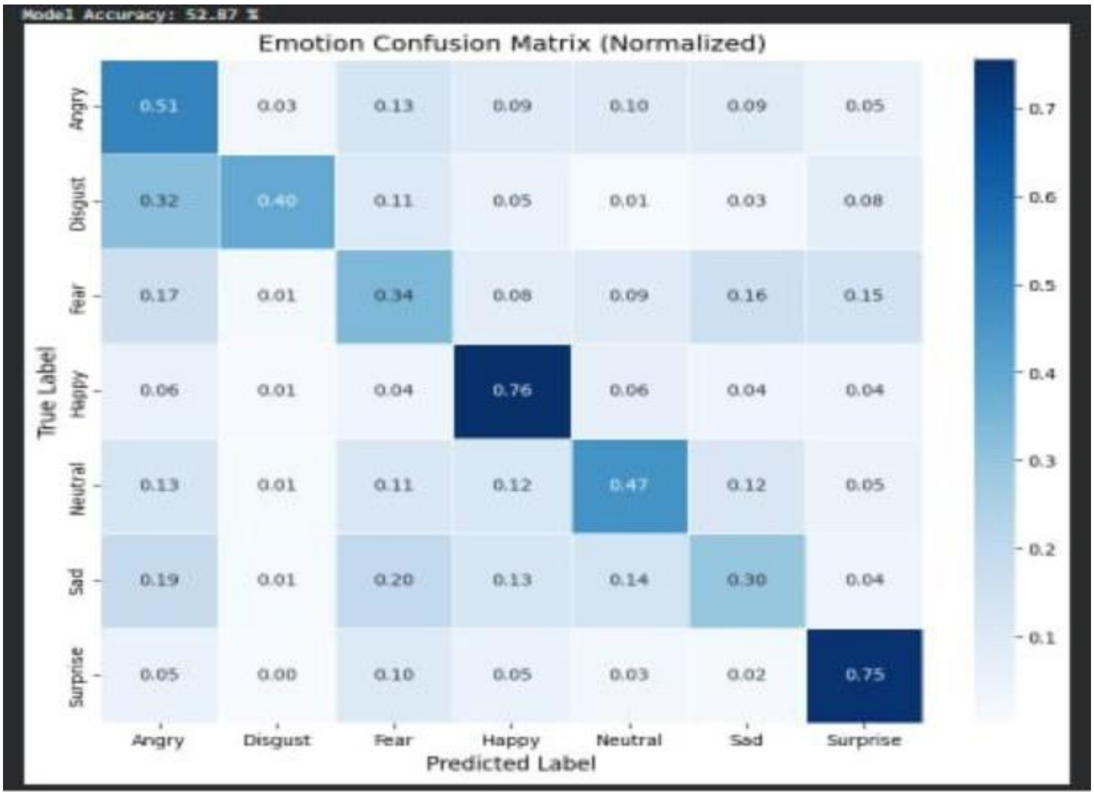
```
              precision    recall  f1-score   support

       Angry       0.39      0.52      0.45       958
     Disgust       0.68      0.29      0.41       111
        Fear       0.38      0.33      0.35      1024
       Happy       0.73      0.79      0.76      1774
     Neutral       0.47      0.54      0.51      1233
         Sad       0.45      0.30      0.36      1247
    Surprise       0.69      0.68      0.69       831

    accuracy                           0.54      7178
   macro avg       0.54      0.49      0.50      7178
weighted avg       0.54      0.54      0.53      7178
```

Generated Results:

Lip Reading Confusion Matrix (Normalized 20 Classes)



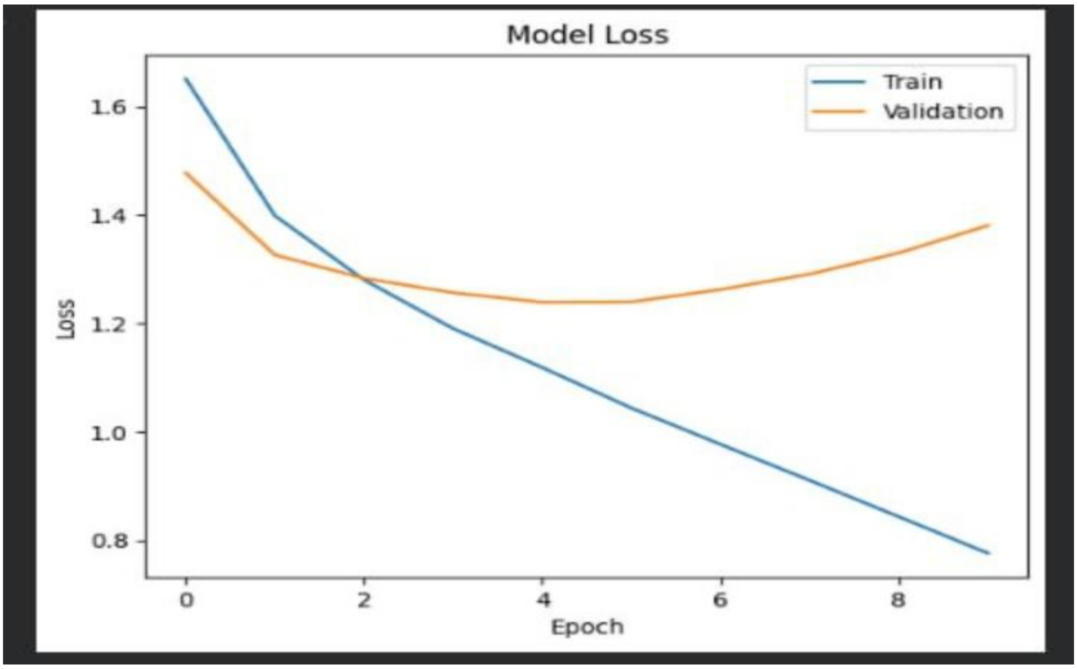
Emotion Confusion Matrix (Normalized)



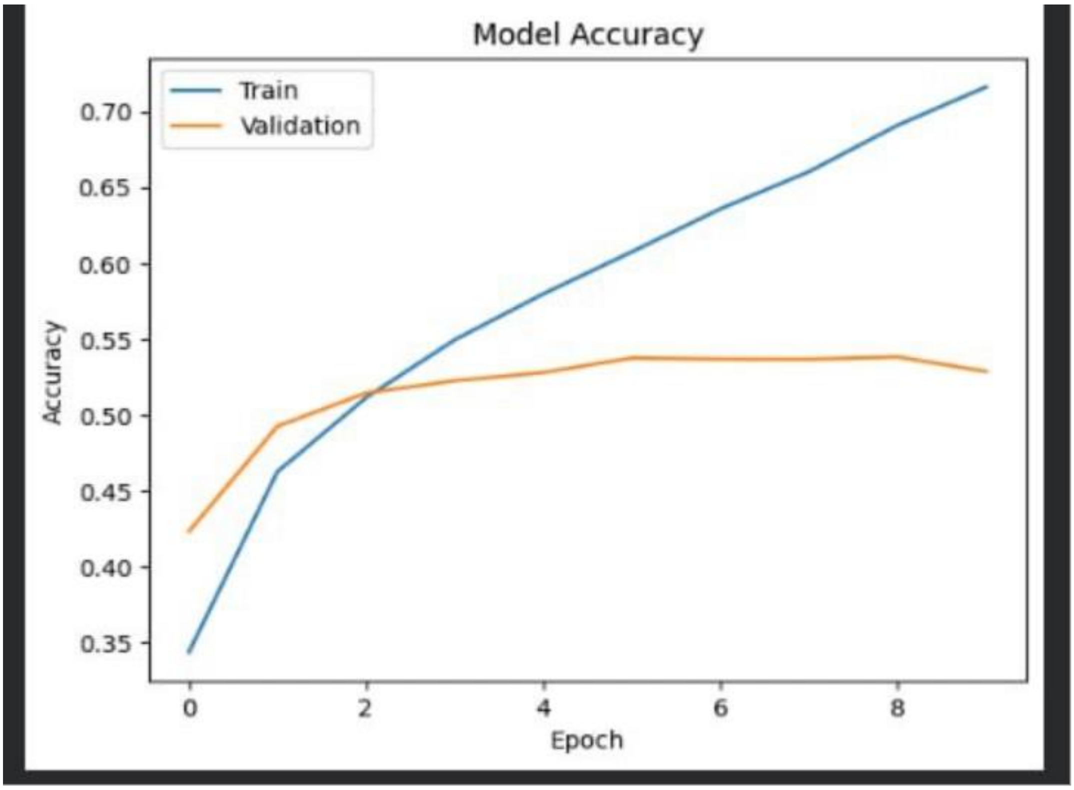
Epoch 10

```
Epoch 1/10
898/898 ━━━━━━━━━━━━━━━━━━━━ 14s 10ms/step - accuracy: 0.3436 - loss: 1.6510 - val_accuracy: 0.4231 - val_loss: 1.4773
Epoch 2/10
898/898 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.4627 - loss: 1.3986 - val_accuracy: 0.4926 - val_loss: 1.3262
Epoch 3/10
898/898 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - accuracy: 0.5114 - loss: 1.2802 - val_accuracy: 0.5145 - val_loss: 1.2829
Epoch 4/10
898/898 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.5495 - loss: 1.1902 - val_accuracy: 0.5226 - val_loss: 1.2564
Epoch 5/10
898/898 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - accuracy: 0.5799 - loss: 1.1183 - val_accuracy: 0.5280 - val_loss: 1.2388
Epoch 6/10
898/898 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.6076 - loss: 1.0436 - val_accuracy: 0.5376 - val_loss: 1.2395
Epoch 7/10
898/898 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.6360 - loss: 0.9765 - val_accuracy: 0.5369 - val_loss: 1.2622
Epoch 8/10
898/898 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - accuracy: 0.6685 - loss: 0.9100 - val_accuracy: 0.5368 - val_loss: 1.2906
Epoch 9/10
898/898 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.6911 - loss: 0.8429 - val_accuracy: 0.5383 - val_loss: 1.3298
Epoch 10/10
898/898 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - accuracy: 0.7164 - loss: 0.7756 - val_accuracy: 0.5287 - val_loss: 1.3804
```

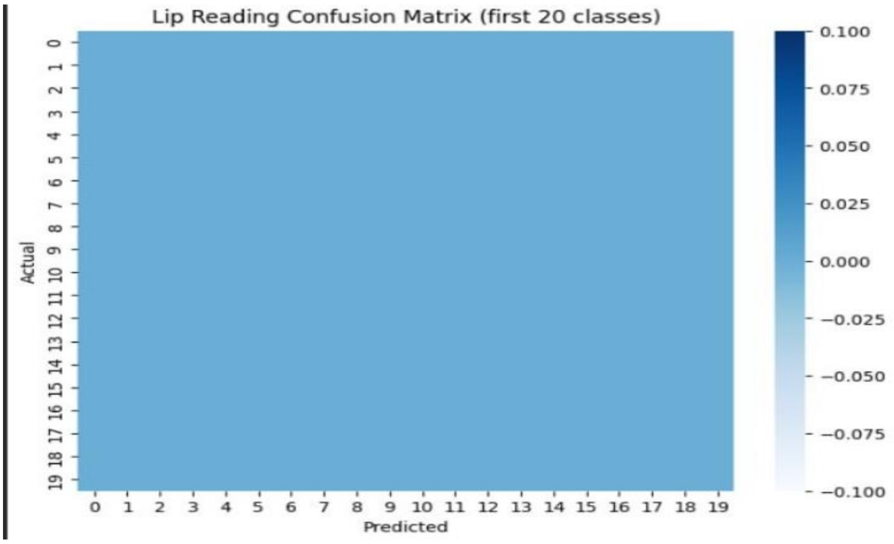
Model Loss



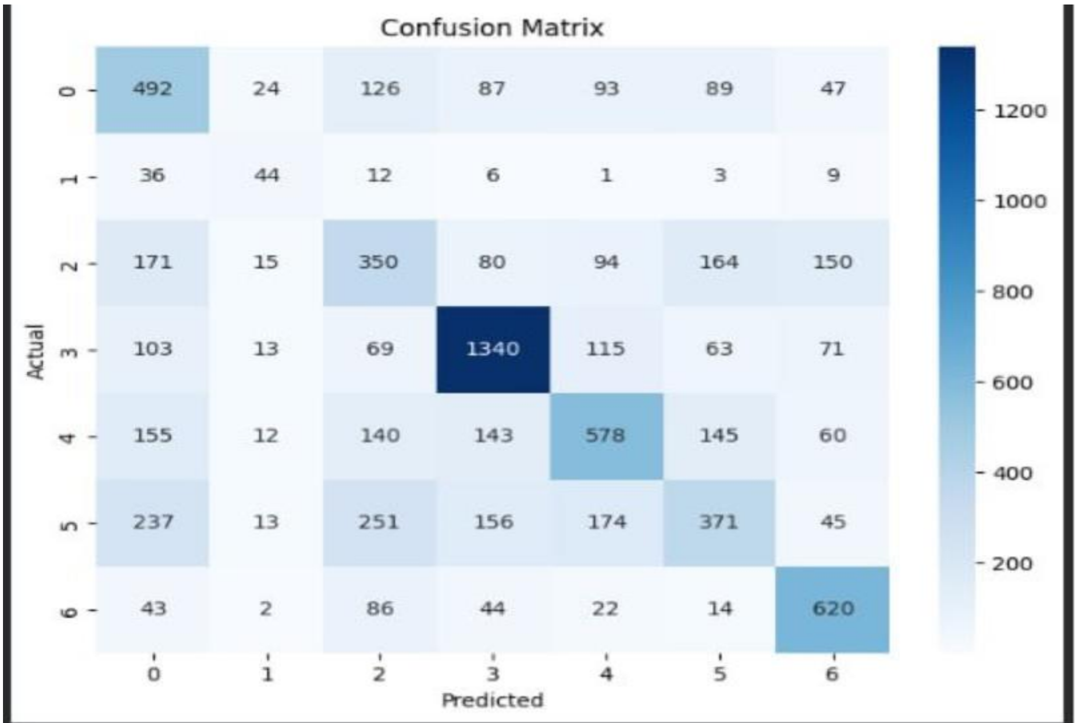
Model Accuracy



Lip Reading Confusion Matrix



Confusion Matrix and Sequential Model CNN




```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/preprocessing/data_layer.py:95: UserWarning: Do not pass an
  super().__init__(**kwargs)
Model: "sequential_4"
```

Layer (type)	Output Shape	Param #
rescaling_4 (Rescaling)	(None, 64, 64, 1)	0
conv2d_12 (Conv2D)	(None, 62, 62, 32)	320
max_pooling2d_12 (MaxPooling2D)	(None, 31, 31, 32)	0
conv2d_13 (Conv2D)	(None, 29, 29, 64)	18,496
max_pooling2d_13 (MaxPooling2D)	(None, 14, 14, 64)	0
conv2d_14 (Conv2D)	(None, 12, 12, 128)	73,856
max_pooling2d_14 (MaxPooling2D)	(None, 6, 6, 128)	0
flatten_4 (Flatten)	(None, 4608)	0
dense_8 (Dense)	(None, 128)	589,952
dense_9 (Dense)	(None, 7)	903

```
Total params: 683,527 (2.61 MB)
Trainable params: 683,527 (2.61 MB)
Non-trainable params: 0 (0.00 B)
```

Conclusion

The system which we propose uses multiple methods to detect emotions through lips while analyzing facial expressions and lip movement. The facial emotion recognition model, which was developed using the FER2013 dataset, achieved moderate success in identifying four fundamental emotions which included happy, sad, angry and neutral. The lip reading system delivered extra visual information which helped researchers analyze how lips moved to better understand speech context.

System performance increased when these two modalities were combined because their joint operation outperformed their separate functions. The system evaluation used accuracy, precision, recall, and F1-score metrics together with confusion matrix analysis to show that the system performs well on clear facial expressions but struggles to identify similar facial emotions. The system operated successfully in a real-time environment which used camera input to detect faces and extract lip regions while predicting emotions with high accuracy. The proposed approach shows its practical use through this demonstration.

The project shows that using facial emotion features together with lip movement data improves system performance because it helps machines understand context better. The method can be extended to develop more sophisticated systems that will enhance user experience in human-computer interaction and assistive technologies and intelligent communication systems.

Reference

1. Ian Goodfellow et al., “Challenges in Representation Learning: A Report on Three Machine Learning Contests,”
   ICONIP, 2013. (FER2013 Dataset)
2. Yannis M. Assael et al., “LipNet: End-to-End Sentence-level Lipreading,” arXiv:1611.01599, 2016.
3. Houwei Cao et al., “CREMA-D: Crowd-sourced Emotional Multimodal Actors Dataset,”
   IEEE Transactions on Affective Computing, 2014.
4. Kaggle, “FER2013 Dataset,”
   https://www.kaggle.com/datasets/msambare/fer2013
5. Kaggle, “Lip Reading Dataset,”
   https://www.kaggle.com/datasets/allenye66/best-lip-reading-dataset
6. Kaggle, “CREMA-D Dataset,”
   https://www.kaggle.com/datasets/ejlok1/cremad
7. Paul Viola and Michael Jones,
   “Rapid Object Detection using a Boosted Cascade,” CVPR, 2001